Attendance Marks Distribution	
% Attendance Obtained	Marks Obtained out of 10%
≥ 95%	10 %
≥ 90% – < 95%	8%
≥ 85% – < 90%	6%
≥ 80% – < 85%	4%
≥ 75% – < 80%	2%
≥ 70% – < 75%	1%
< 70%	Not eligible for the final exam.

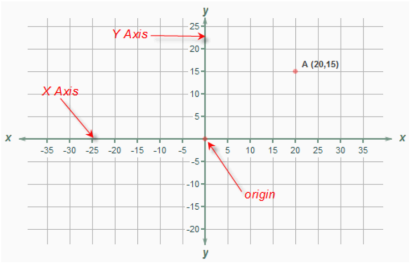
- **No excuses will be accepted for absence from class**.
- Late Attendance: A maximum delay of 15 minutes may be considered, provided that the student is able to attend the next class on time.

CO No.	Course Outcomes (CO) Statement
CO1	Understand and apply the concepts of 2D and 3D coordinate geometry, including transformation of axes, pair of straight lines, circles, planes, straight lines, spheres, conicoids, cones, and cylinders, to solve geometric problems.
CO2	Understand and apply the concepts and techniques of differential calculus, including limits, continuity, differentiation, higher-order derivatives, Taylor's and Maclaurin's theorems, partial differentiation, extrema, curvature, asymptotes, envelopes, curve tracing, and related applications.

Mapping of Course Outcomes (COs) and Evaluation Methods

Assessment Method	Marks	CO1	CO2	Final
		PO1	PO1	
Attendance (Class Participation)	10% (40)	Eligibility to sit for the final exam of the course requires 70% attendance.		
Quiz (I-IV)/Assignment/ H.W.	30% (120)	15%	15%	As obtained (Best four)
Midterm Exam.	30% (120)	15%	15%	As obtained
Final Exam.	30% (120)	15%	15%	As obtained
Total	100% (400)	45%	45%	As obtained

Basic:



Definition of plane coordinate: Plane coordinate geometry is that branch of geometry in which we use two numbers called coordinates to indicate the position of a point in a plane.

Abscissa: The distance from the y-axis is called the X co-ordinate or abscissa of the point.

Ordinate: The distance from the X-axis is called the Y co-ordinate or ordinate of the point.

Coordinate: The two distances taken together are called the co-ordinates of the point & are represented by the symbol (x,y)

Change of Axes: (Try yourself)

1. Determine the equation of the curve $x^2 + 2y^2 - 6x + 7 = 0$ when the origin is transferred to the point (3,1).
2. Determine the equation of the curve $x^2 + y^2 - x + 7 = 0$ when the origin is transferred to the point (1,1).
3. Transform the equation $x^2 + 2\sqrt{3}xy - y^2 = 2a^2$ after rotating of axes through an angle$30°$.
4. Transform the equation $x^2 + xy - y^2 = 0$ after rotating of axes through an angle$45°$.
5. Remove the first degree terms of the given equation $36x^2 + 24xy + 29y^2 - 72x + 126y + 81 = 0$.
6. Remove the first degree terms of the given equation $3x^2 - 4y^2 - 6x - 8y - 10 = 0$.
7. Remove the first degree terms of the equation.$6x^2 + 4xy + 29y^2 - 72x + 12y + 90 = 0$.
8. Determine the equation of the curve $2x^2 + 3y^2 - 8x + 6y - 7 = 0$, when the origin is transferred to the point (2,3) ,where the direction of the axes remains unaltered.
9. Determine the equation of the curve $3x^2 + 14xy - 24y^2 - 22x + 110y - 121 = 0$, when the origin is transferred to the point (-1,2) ,where the direction of the axes remains unaltered.
10. Determine the equation of the curve $2x^2 + y^2 - 4x + 4y = 0$, when the origin is transferred to the point (1,-2) ,where the direction of the axes remains unaltered.
11. Determine the equation of the curve $2x^2 + 3y^2 - 8x + 6y - 7 = 0$, when the origin is transferred to the point (-2,-1) ,where the direction of the axes remains unaltered.
12. Remove the first degree terms of the equation $x^2 + 24xy + y^2 + 2x + 12y - 20 = 0$.
13. Remove the first degree terms of the equation $5x^2 + 12xy + 20y^2 - 12x + 16y + 50 = 0$.

Thanks for your attention